

David Attard

☎ (+44) 7407165286
✉ davidattard4@gmail.com
📄 david-attard.github.io
ORCID: 0009-0006-4560-5486

Education

- 2023–2027 **PhD in Astronomy**, *University of Sussex*, Brighton, United Kingdom.
(Expected) Investigating the impact of the Epoch of Reionization on early galaxy formation.
Supervisor: Prof. Ilian T. Iliev
- 2019–2023 **B.Sc. (Hons) Mathematics and Physics**, *University of Malta*, Msida, Malta, First Class Honours, Average mark: 92%.
Included a one-year ERASMUS+ exchange (2021–2022) at Cardiff University, attending the Schools of Mathematics and Physics and Astronomy.

Research Experience

- Sep 2023–Present **Postgraduate Researcher**, UNIVERSITY OF SUSSEX, United Kingdom, Supervisor: Prof. Ilian T. Iliev.
- Investigating how radiative feedback during the Epoch of Reionization regulates the formation and evolution of the first galaxies, using cosmological simulations and radiative-transfer modelling.
 - Led a study of the reionization history of the Local Group using the constrained HESTIA simulations, determining the timing and spatial progression of reionization around Milky Way and M31 analogues.
 - Examining how alternative stellar-feedback prescriptions affect dwarf-galaxy quenching in the MEGATRON radiation-hydrodynamic simulations, with particular emphasis on the relationship between early assembly, feedback mechanisms, and long-term quiescence.
 - Developing Lyman–Werner radiative-transfer capabilities in pyC2Ray to quantify the contribution of minihaloes and Population III stars to early radiation backgrounds and the suppression of molecular cooling.
- Jul 2023–Aug 2023 **CERN Summer Student**, CERN, Geneva, Switzerland.
- Developed and benchmarked machine-learning inference approaches for FastCaloGAN, comparing ONNX Runtime and ROOT/TMVA SOFIE for fast detector simulation.
 - Evaluated the speed, memory usage, and accuracy of reduced-precision models across CPU and GPU architectures.

Professional Experience

- Sep 2025–Nov 2025 **Data Science Intern**, *Penta Group*, London, United Kingdom, DISCNet- industry placement.
- Built and deployed NLP pipelines using named entity recognition to extract and disambiguate entities from unstructured, multi-terabyte text corpora.
 - Designed and tuned programmatic LLM prompting strategies for large-scale classification tasks, evaluating output quality against manually labelled validation sets.
 - Applied scikit-learn for feature engineering, clustering and model evaluation within the classification workflow.
 - Delivered an end-to-end influence-identification tool from prototype to production, including data curation, query optimisation and handover to non-technical users.

Technical Skills

- Programming Python, Fortran 90, Bash, Git
- HPC & GPU CUDA, MPI, SLURM, distributed HPC workflows, GPU-accelerated code development, profiling and optimisation
- Scientific Computing NumPy, SciPy, Pandas, Pynbody, pyc2ray
- Languages English (fluent), Maltese (native), German (beginner)

Publications

Refereed Journal Articles

- 2026 **Attard, D.**, Conaboy, L., Libeskind, N., Pillipenko, S., Dixon, K., & Iliev, I. T.
“Reionization in HESTIA: studying reionization in the Local Group through zoom simulations.”
Monthly Notices of the Royal Astronomical Society, 547(3), stag467. doi:10.1093/mnras/stag467

- 2025 Iliev, I. T., Hosein, A. R., Chluba, J., Conaboy, L., **Attard, D.**, Mondal, R., Ahn, K., Gottlöber, S., Lewis, J., Ocvirk, P., Park, H., Shapiro, P. R., Sorce, J. G., & Yepes, G.
"The thermal Sunyaev–Zel'dovich effect from the Epoch of Reionization."
Monthly Notices of the Royal Astronomical Society, 540(2), 1432. doi:10.1093/mnras/staf768

Manuscripts in Preparation

- In prep. **Attard, D.**, Iliev, I.T., Rey, M. P., Katz, H., Cadiou, C., Choustikov, N.
"Galaxy quenching in the Epoch of Reionization with the MEGATRON simulation suite."
- In prep. **Attard, D.** et al. Lyman-Werner radiative feedback from Pop III sources and minihalos in pyc2ray.

Grants

Successful grant applications which I have contributed to:

Contributor

- 2025 JUWELS (Jülich, Germany) computing time (PI: Dr Noam Libeskind, PC: Prof Ilian Iliev) 14.2M core-h, £213k
- 2024 JUWELS (Jülich, Germany) computing time (PI: Dr Noam Libeskind, PC: Prof Ilian Iliev) 8.8M core-h, £132k

Talks and Presentations

- Sep 2026 **BUGS II.**
Poster presented titled "MEGATRON: Quenching of Dwarf Galaxies in the EoR"
- June 2026 **CLUES Meeting.**
Presented the talk Understanding Galaxy Formation in the Epoch of Reionization.
- Dec 2025 **DISCNet Showcase.**
Presented work on adapting astrophysical data analysis skillsets to industry-scale data infrastructure solutions.
- June 2025 **CLUES Meeting.**
Contributed talk on Local Group reionization using constrained zoom simulations.
- June 2024 **CLUES Meeting.**
Contributed talk presenting radiative transfer post-processing results on the HESTIA simulation suite.
- 2024–2026 **Astronomy Group Seminar, University of Sussex, Brighton, United Kingdom.**
Research presentations on Local Group reionization and radiative-transfer simulations.

Teaching and Academic Service

- Jan **Doctoral Tutor, University of Sussex, United Kingdom.**
- 2024–May 2026 Led interactive computational/analytical workshops, evaluated assignments, and provided pedagogical support for undergraduate courses:
- Foundational Mechanics
 - Foundational Mathematics
 - Mathematical Methods 3
- Sep 2024–Sep 2025 **Journal Club Organiser, University of Sussex, United Kingdom.**
Coordinated journal club meetings, scheduled presenters, and facilitated discussions of recent astrophysics research.
- Nov 2025 **STEM Outreach Volunteer, Lewes STEM Festival, United Kingdom.**
Represented the University of Sussex Astronomy Department, engaging members of the public with astronomy activities and scientific demonstrations.

Scholarships and Awards

- 2024 **Evolve Special Prize for Science, University of Malta.**
- 2023 **DISCNet PhD Scholarship, University of Sussex.**
- 2019–2022 **Dean's List, Faculty of Science, University of Malta.**

References

Prof. Ilian Iliev, Department of Physics & Astronomy, University of Sussex, Brighton, BN1 9QH, UK, i.t.iliev@sussex.ac.uk

Dr. Martin Rey, Department of Physics, University of Bath, Claverton Down, Bath BA2 7AY, UK, mpr47@bath.ac.uk

Dr. Noam Libeskind, Leibniz-Institut für Astrophysik Potsdam (AIP), An der Sternwarte 16, D-14482 Potsdam, Germany, nlibeskind@aip.de